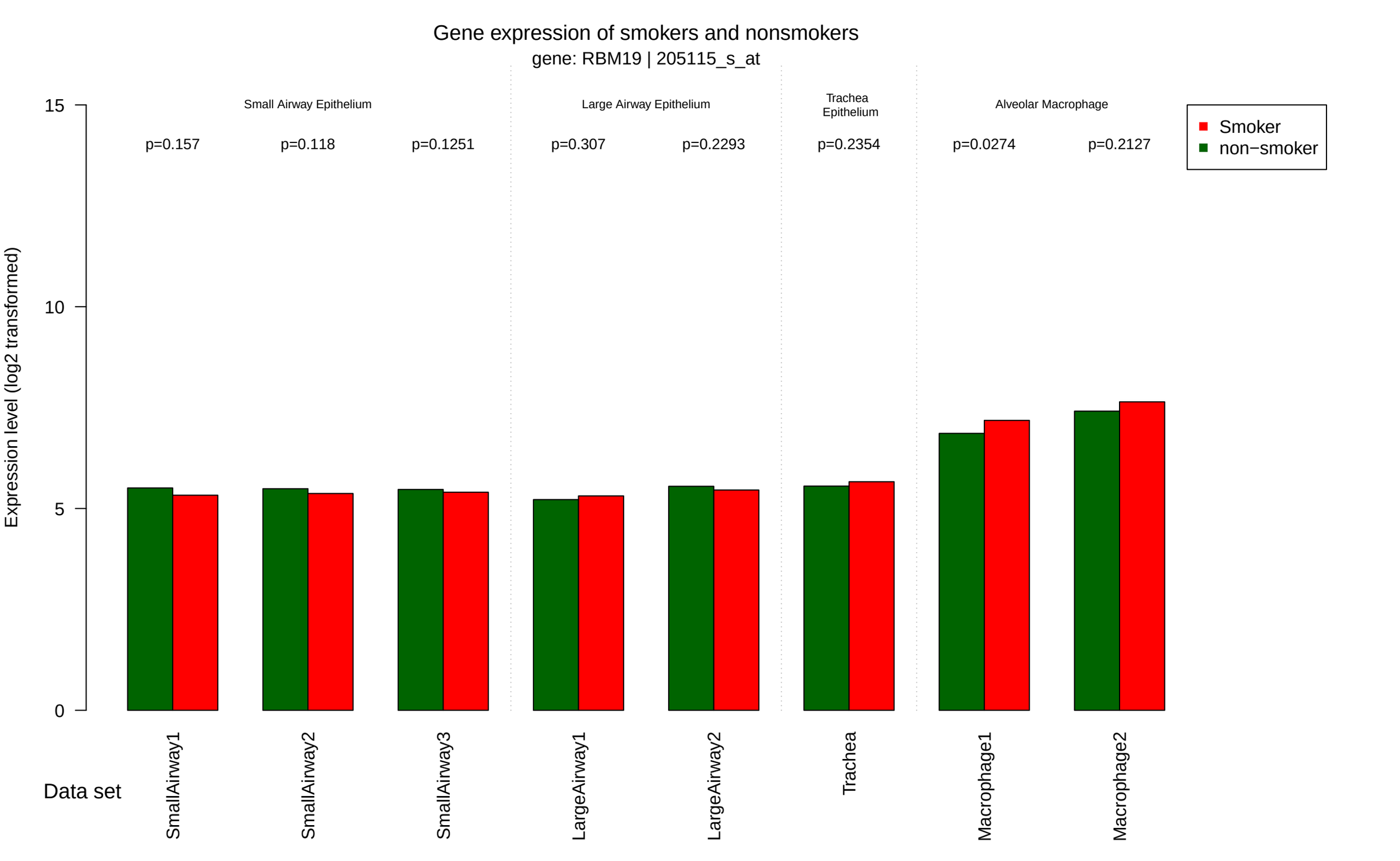
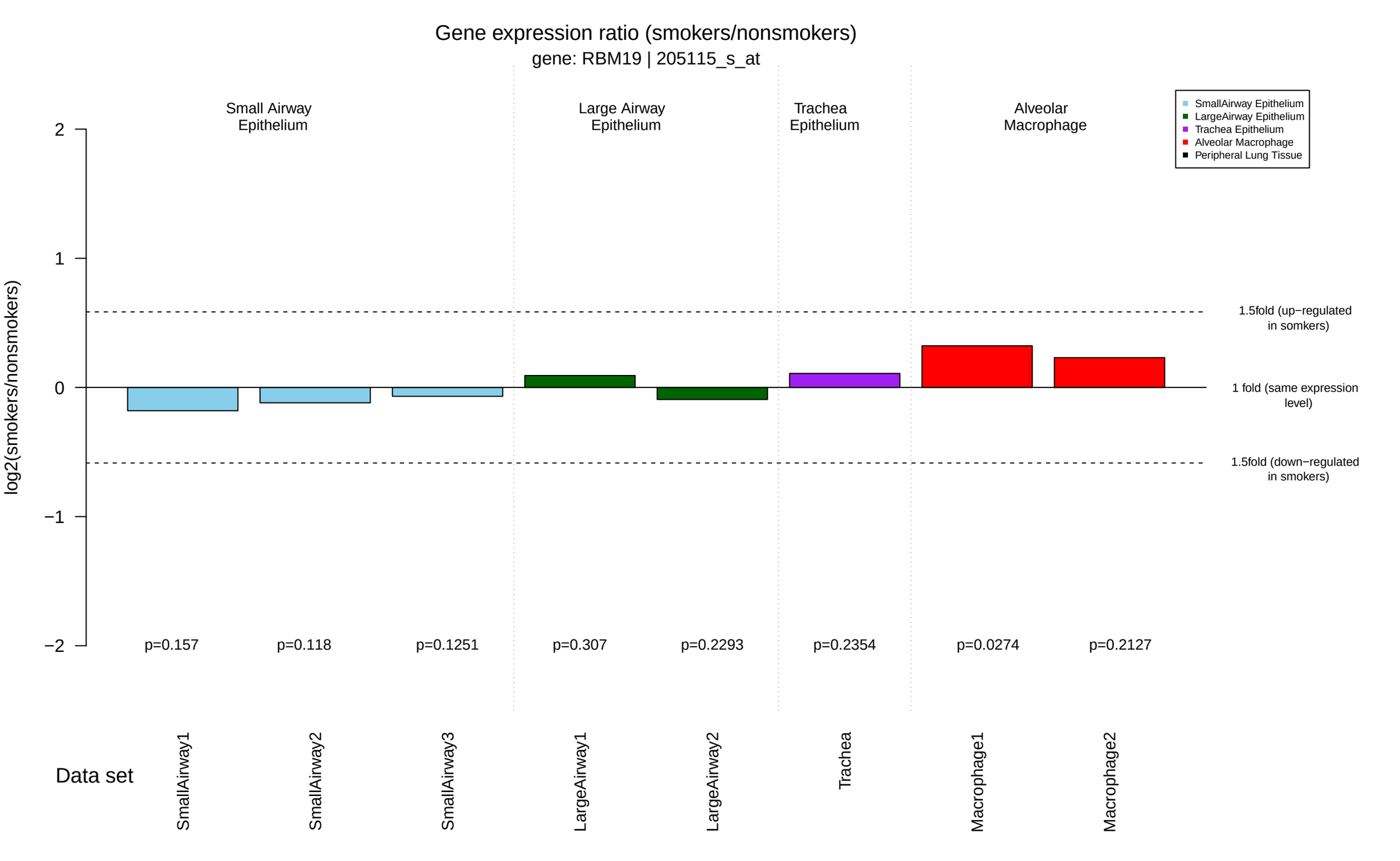
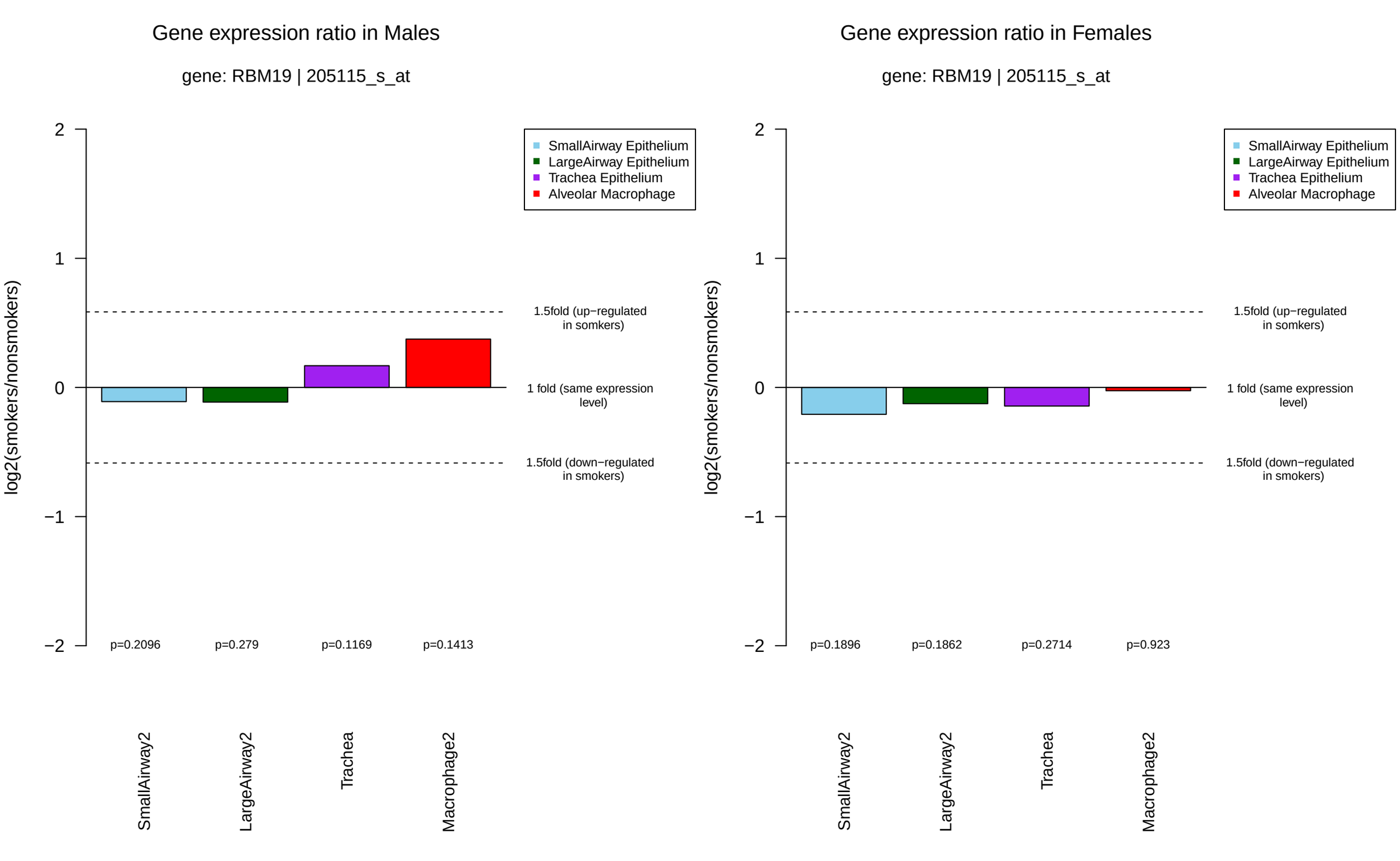
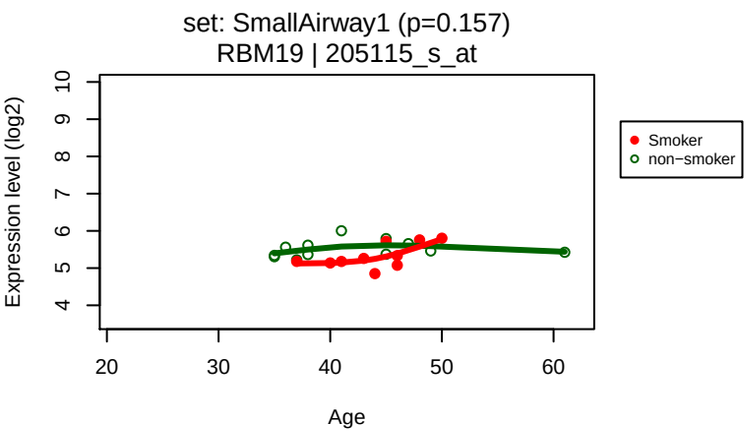
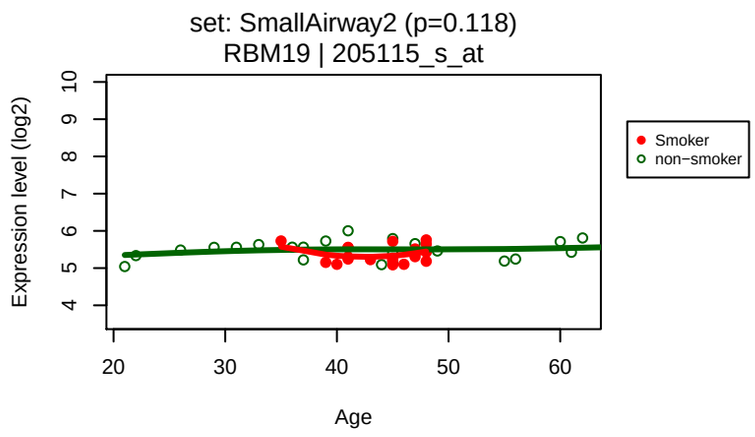
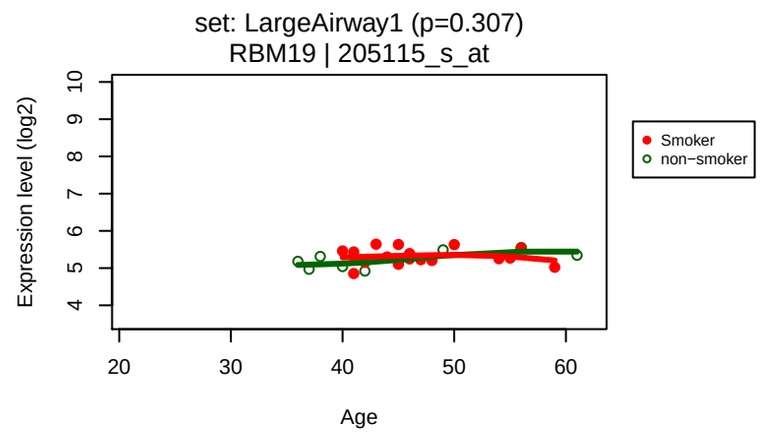

Gene expression of smokers and nonsmokers
gene: RBM19 | 205115_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.157
p=0.118
p=0.1251
p=0.307
p=0.2293
p=0.2354
p=0.0274
p=0.2127
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

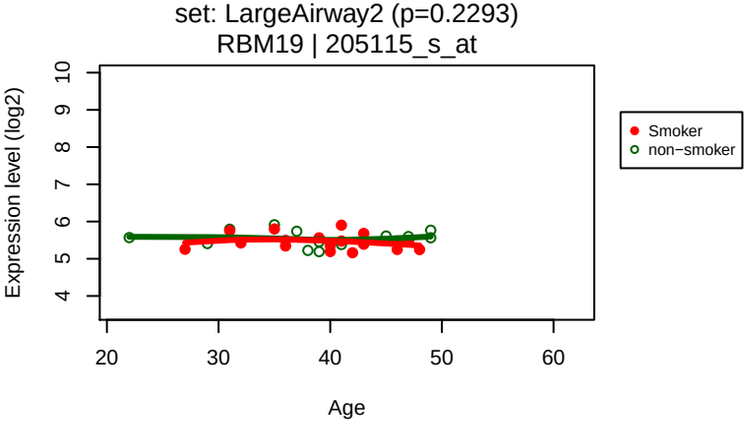

Gene expression ratio (smokers/nonsmokers)
gene: RBM19 | 205115_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.157
p=0.118
p=0.1251
p=0.307
p=0.2293
p=0.2354
p=0.0274
p=0.2127
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

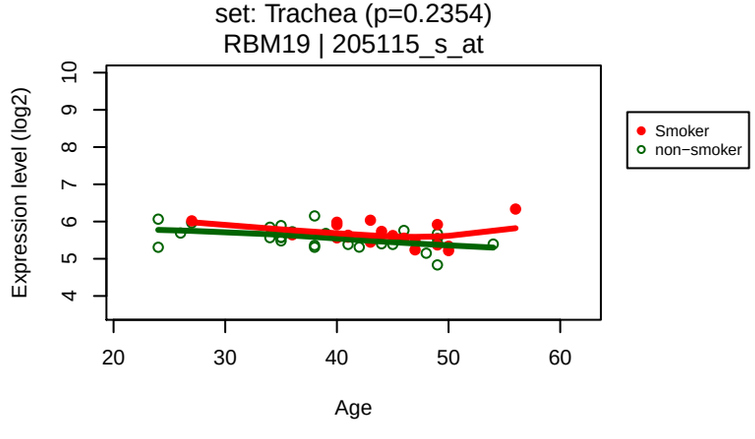

Gene expression ratio in Males
gene: RBM19 | 205115_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2096
p=0.279
p=0.1169
p=0.1413
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RBM19 | 205115_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1896
p=0.1862
p=0.2714
p=0.923
SmallAirway2
LargeAirway2
Trachea
Macrophage2

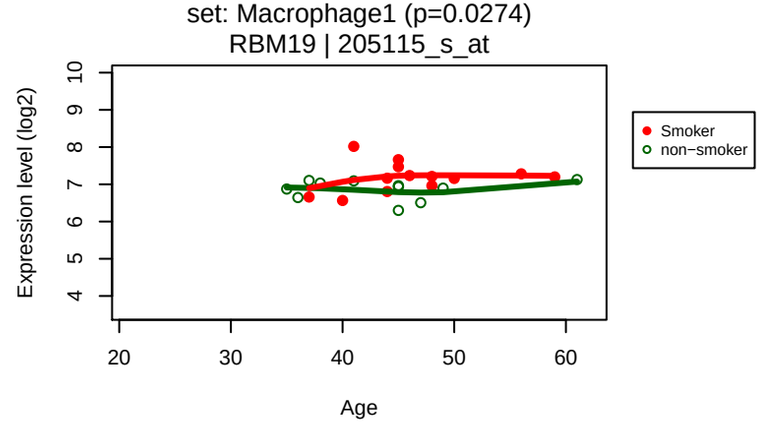

set: SmallAirway1 (p=0.157)
RBM19 | 205115_s_at
Expression level (log2)
Age
Smoker
non−smoker

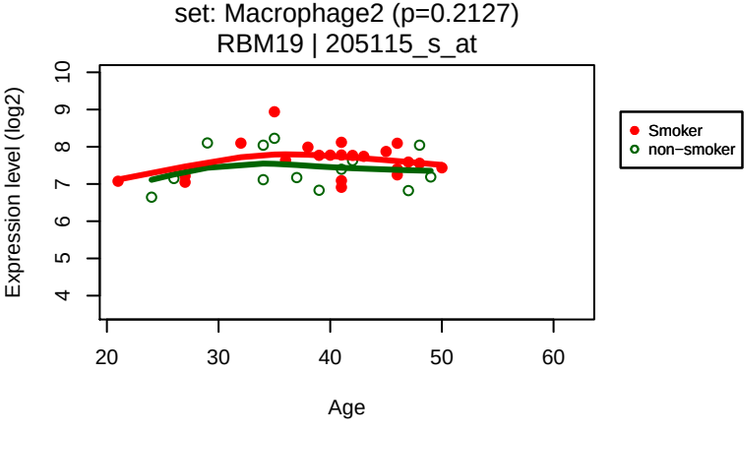

set: SmallAirway2 (p=0.118)
RBM19 | 205115_s_at
Expression level (log2)
Age
Smoker
non−smoker

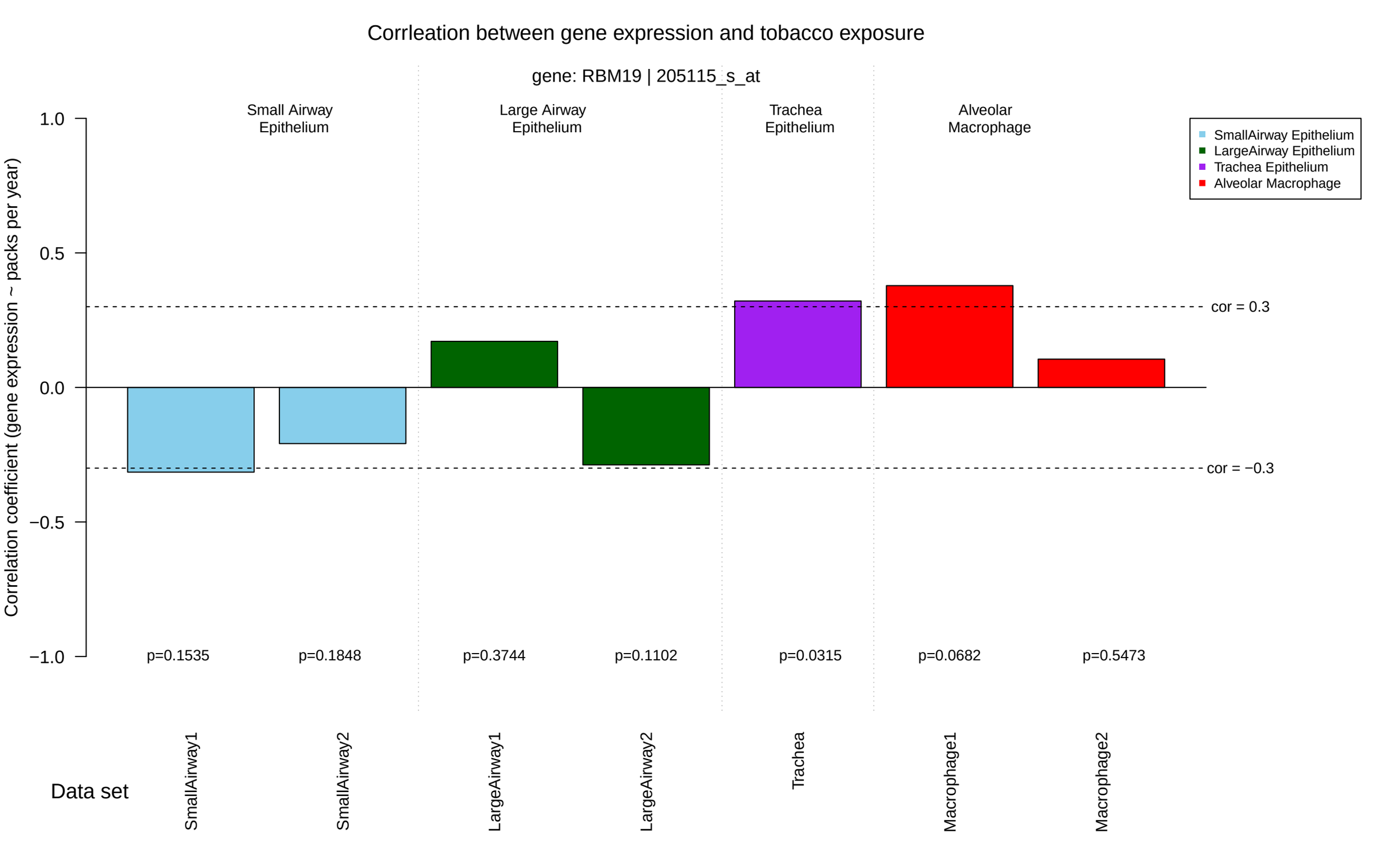

set: LargeAirway1 (p=0.307)
RBM19 | 205115_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2293)
RBM19 | 205115_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2354)
RBM19 | 205115_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0274)
RBM19 | 205115_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2127)
RBM19 | 205115_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RBM19 | 205115_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1535
p=0.1848
p=0.3744
p=0.1102
p=0.0315
p=0.0682
p=0.5473
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2